$$B_H = \frac{\mu_\circ N}{2R}\frac{I}{\tan\theta} \qquad (3.41)$$

EXAMPLE 3.14

A coil of a tangent galvanometer of diameter 0.24 m has 100 turns. If the horizontal component of Earth's magnetic field is 25×10^{-6} T then, calculate the current which gives a deflection of 60°.

Solution

The diameter of the coil is 0.24 m. Therefore, radius of the coil is 0.12 m.

Number of turns is 100 turns.

Earth's magnetic field is 25×10^{-6} T

Deflection is

$$\theta = 60^\circ \Rightarrow \tan 60^\circ = \sqrt{3} = 1.732$$

$$I = \frac{2RB_H}{\mu_\circ N}\tan\theta$$

$$= \frac{2\times 0.12\times 25\times 10^{-6}}{4\times 10^{-7}\times 3.14\times 100}\times 1.732 = 0.82\times 10^{-1}\text{A}.$$

$$I = 0.082 \text{ A}$$

3.8.5 Current loop as a magnetic dipole

The magnetic field at a point on the axis of the current-carrying circular loop of radius R at a distance z from its centre is given by

$$\vec{B} = \frac{\mu_\circ I}{2}\frac{R^2}{\left(R^2+z^2\right)^{\frac{3}{2}}}\hat{k} \quad \text{(From eqn. 3.37)}$$

At larger distance $z >> R$, therefore $R^2 + z^2 \approx z^2$, we have

$$\vec{B} = \frac{\mu_\circ I}{2}\frac{R^2}{z^3}\hat{k} \text{ or } \vec{B} = \frac{\mu_\circ I}{2\pi}\frac{\pi R^2}{z^3}\hat{k} \qquad (3.42)$$

Let A be the area of the circular loop $A = \pi R^2$. So rewriting the equation (3.42) in terms of area of the loop, we have

$$\vec{B} = \frac{\mu_\circ I}{2\pi}\frac{A}{z^3}\hat{k}$$

(or)

$$\vec{B} = \frac{\mu_\circ}{4\pi}\frac{2IA}{z^3}\hat{k} \qquad (3.43)$$

Comparing equation (3.43) with equation (3.14) dimensionally, we get

$$p_m = I\,A$$

where p_m is called magnetic dipole moment. In vector notation,

$$\vec{p}_m = I\,\vec{A} \qquad (3.44)$$

This implies that a current carrying circular loop behaves as a magnetic dipole of magnetic moment $\vec{p}_m$. So, **the magnetic dipole moment of any current loop is equal to the product of the current and area of the loop.**

Right hand thumb rule

In order to determine the direction of magnetic moment, we use right hand thumb rule which states that

If we curl the fingers of right hand in the direction of current in the loop, then the stretched thumb gives the direction of the magnetic moment associated with the loop.

Table 3.3 End rule – polarity with direction of current in circular loop

Current in circular loop	Polarity	Picture
Anti-clockwise current	North Pole	 Anti-clockwise current Polarity: North Pole
Clockwise current	South Pole	 Clockwise current Polarity: South Pole

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